THE PONTRYAGIN-STYLE DUAL MODULE OF A MODULE IS INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Pontryagin dual of a module with respect to the dual of the integers and the module of homomorphisms into it. discussed in the same block, source lines 13794-13827. Supplies the definition of the dualising module and the adjunction identity used in the proof.
- Products of injective modules are injective. used at proof line 13836. Reduces the statement to injectivity of the dual of the ring.
- The duality functor is exact. the author cites Lemma lemma-vee-exact at proof lines 13843-13845. Gives injectivity of the dual of the ring, hence of the product.

Source reference. Original lemma 01DC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1793–1793. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01DA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1791–1791. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. For every R-module M the R-module $J(M)$ is injective.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Note that $J(M) \cong \prod_{\varphi \in M^\vee} R^\vee$ as an R-module. As the product of injective modules is injective, it suffices to show that $R^\vee$ is injective. For this we use that

$$\mathrm{Hom}_R(N, R^\vee) = \mathrm{Hom}_R(N, \mathrm{Hom}_{\mathbf{Z}}(R, \mathbf{Q}/\mathbf{Z})) = N^\vee$$

and the fact that $(-)^\vee$ is an exact functor by Lemma 01DA. $\square$